Stop Shuffling Ballot Papers

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Australian electoral innovations

- Secret ballot ("The Australian Vote")
- Compulsory voting
- Compulsory registration
- Preferential voting (AV)
- Single transferable vote (STV) (Hare-Clark)
- Gregory method
- Saturday voting
- The pencil

The story, in a sentence

Moral

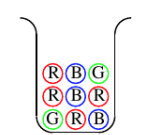
Counting votes in complicated voting systems is hard if we focus on the ballot papers. . .
Everything is simplified if we focus on the multiset of vote types

(Get the datatypes right!)

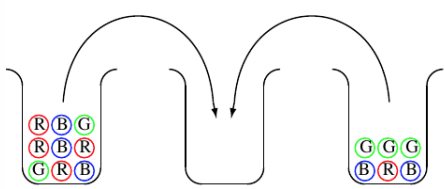




Multisets and their properties



$4|R\rangle + 3|B\rangle + 2|G\rangle$



1. Multisets over X form a commutative monoid
2. Write $\mathcal{L}$ for the List functor **set** $\to$ **set** (as usual), and analagously then
3. $\mathcal{N}$, sending a set X to $\mathcal{N}(X)$, the set of multisets over X, is a functor
4. $\mathcal{N}$ is in fact a monad
5. Accumulation of lists into multisets is natural, ie, $acc : \mathcal{L}(X) \to \mathcal{N}(X)$ satisfies $acc \circ \mathcal{L}(f) = \mathcal{N}(f) \circ acc$, for each function f

Counting votes – referenda and instant run-off

Referendum (usually just "yes or no") Just acc and Max

First past the post (eg Makerfield/Holborn – who has the most votes) Just acc and Max

Instant run-off (also called preferential voting)

- Make first preference piles then
- If Max $\leq 50\%$
- Identify Min
- Redistribute from Min to next available preference
- Repeat



Instant run-off via multisets

All multiset algorithms begin with acc to turn the sequence of ballot papers into a multiset of votes (in the instant run-off case, a list of lists into a multiset of lists), then

- $\mathcal{N}(hd)$ is equivalent to making the first preference piles
- If Max $\leq 50\%$
- Identify $x =$ Min
- $\mathcal{N}(del_x)$
- Repeat

Instant run-off example

	7	2	7	8	7	19	12	12	8	6	12
1st	a	a	a	b	b	b	c	d	d	d	d
2nd	b	c	d	a	c	d	a	a	a	b	c
3rd	d	d	c	c	d	c	d	b	c	c	a
4th	c	b	b	d	a	a	b	c	b	a	b

(1)

$$\begin{aligned}\varphi_0 \;=\; & 7\big|\,a,b,d,c\,\big\rangle + 2\big|\,a,c,d,b\,\big\rangle + 7\big|\,a,d,c,b\,\big\rangle + 8\big|\,b,a,c,d\,\big\rangle \\ & + 7\big|\,b,c,d,a\,\big\rangle + 19\big|\,b,d,c,a\,\big\rangle + 12\big|\,c,a,d,b\,\big\rangle + 12\big|\,d,a,b,c\,\big\rangle \\ & + 8\big|\,d,a,c,b\,\big\rangle + 6\big|\,d,b,c,a\,\big\rangle + 12\big|\,d,c,a,b\,\big\rangle.\end{aligned}$$

$$\begin{aligned}\mathcal{N}\big(hd\big)(\varphi_0) \;&=\; 7\big|\,a\,\big\rangle + 2\big|\,a\,\big\rangle + 7\big|\,a\,\big\rangle + 8\big|\,b\,\big\rangle + 7\big|\,b\,\big\rangle + 19\big|\,b\,\big\rangle + 12\big|\,c\,\big\rangle + 12\big|\,d\,\big\rangle + \ldots \\ &=\; 16\big|\,a\,\big\rangle + 34\big|\,b\,\big\rangle + 12\big|\,c\,\big\rangle + 38\big|\,d\,\big\rangle\end{aligned}$$

Instant run-off — delete c and then a

$$
\begin{aligned}
\varphi_1 &= \mathcal{N}(del_c)(\varphi_0) \\
&= 7|a,b,d\rangle + 2|a,d,b\rangle + 7|a,d,b\rangle + 8|b,a,d\rangle + 7|b,d,a\rangle + 19|b,d,a\rangle \\
&\quad + 12|a,d,b\rangle + 12|d,a,b\rangle + 8|d,a,b\rangle + 6|d,b,a\rangle + 12|d,a,b\rangle. \\
&= 7|a,b,d\rangle + 21|a,d,b\rangle + 8|b,a,d\rangle + 26|b,d,a\rangle + 32|d,a,b\rangle + 6|d,b,a\rangle.
\end{aligned}
$$

$$\mathcal{N}(hd)(\varphi_1) = 28|a\rangle + 34|b\rangle + 38|d\rangle$$

$$\varphi_2 = \mathcal{N}(del_a)(\varphi_1) = 41|b,d\rangle + 59|d,b\rangle$$

$$\mathcal{N}(hd)(\varphi_2) = 41|b\rangle + 59|d\rangle$$

Multi-member electorates and single transferable votes (STV)

- ACT and Tasmania lower houses
- Australian Senate, SA and Victoria upper houses
- NSW upper house
- Western Australian upper house
- Ireland Lower House
- Malta

MME-STV is a common approach to "proportional representation".

The "quota" is roughly V/S, or even better (Droop style), $qu(\varphi, S) = \left\lfloor \frac{\|\varphi\|}{S+1} \right\rfloor + 1$.

What do we do with the surplus?

How to transfer the surplus

- Surplus votes (last to arrive)
- Random selection
- Gregory style – fractonal (!)
- And do you multiply fractions? (!!)
- Last parcel Hare-Clark
- The thousand vote vote

Spoiler: Do we "transfer the surplus" or do we "devalue the votes"?
$(K - Q)/K$ is the transfer fraction, but instead of transferring votes at that fraction we can simply devalue the votes by Q/K and delete (after declaring elected) candidate k.

The MME-STV process in multiset form

We need to allow rational multiplicities, so extend the multiset functor $\mathcal{N}$ to $\mathcal{M}$.
As promised, all multiset algorithms begin with acc to turn the sequence of ballot papers into a (now $\mathcal{M}$) multiset φ_0 of votes (in this case too, a multiset of lists), then

- $\mathcal{M}(hd)$ is equivalent to making the first preference piles
- If Max $\leq Q$
- Identify $x =$ Min
- $\mathcal{M}(del_x)$
- Try again
- If Max $> Q$ choose a candidate, say k, with $K > Q$ votes, declare k elected and
- $\varphi_{i+1} = \mathcal{M}del_k(\varphi_i - (Q/K)\{\varphi_i | hd = k\})$

Proving properties

There should be a naturality result that says if k_1 and k_2 both have greater than Q votes then they can be elected in either order or concurrently without affecting the outcome. There isn't. This has led to

- Wright
- Meek

Wright is remarkably easy to describe in multiset terms.
Relating Wright and Meek in multiset terms is on-going work.
Likewise, sharing the simplifications with electoral authorities is on-going work.
Multisets work well for other systems too, including de Borda and d'Hondt.

Summary

Mathematicians have been deeply involved in proposing and implementing electoral systems

- Condorcet
- de Borda
- Droop
- Gregory
- Meek
- van der Craats (actually a systems analyst) – the inventor of Wright

but they, like the electoral authorities, focused largely on shuffling ballot papers, resulting in long, detailed, complicated algorithms and opaque rules of process, some of which are mathematically necessary, and some of which merely avoid ambiguity in process. Focusing instead on the multisets of votes gives short simple algorithms that can provide reference implementations and about which proofs of properties are much more tractable. And they correspond to the public's intuitions of how the system should work.